Mini Project in Java
Assignment 02

Problem Statement

Write a program in Java showing hierarchial inheritence with base class as employee and derived class as Full Time Employee with methods Display Salary in base class and Calculate salary in derived classes. Calculate salary method will calculate as per inerest given to fulltime and intern employees. Fulltime employee - 50% hike, Intern employee - 25% hike. Display salary before and after hike

Objective:

- To study inheritance in Java
- To study why to use inheritance.
- To study type of inheritance.

Theory:

1) What is Inheritance in Java

Inheritance is an important features of OOC in Java. It allows one class to acquire the properties and methods of another class (Parent or superclass). The keywords extends is used to achieve inheritence in Java.
Inheritance provides code reusability, reduces redundancy, and improves program stucture.

(2) Why use inheritance?

- It helps in reusing existing code.
- It reduces duplication of code.
- It improves readability and maintainability.
- It supports method overriding.
- It represents IS-A ~~employee~~ relationship

eg; Full Time Employee IS-A Employee.

(3) Types of inheritance in Java.

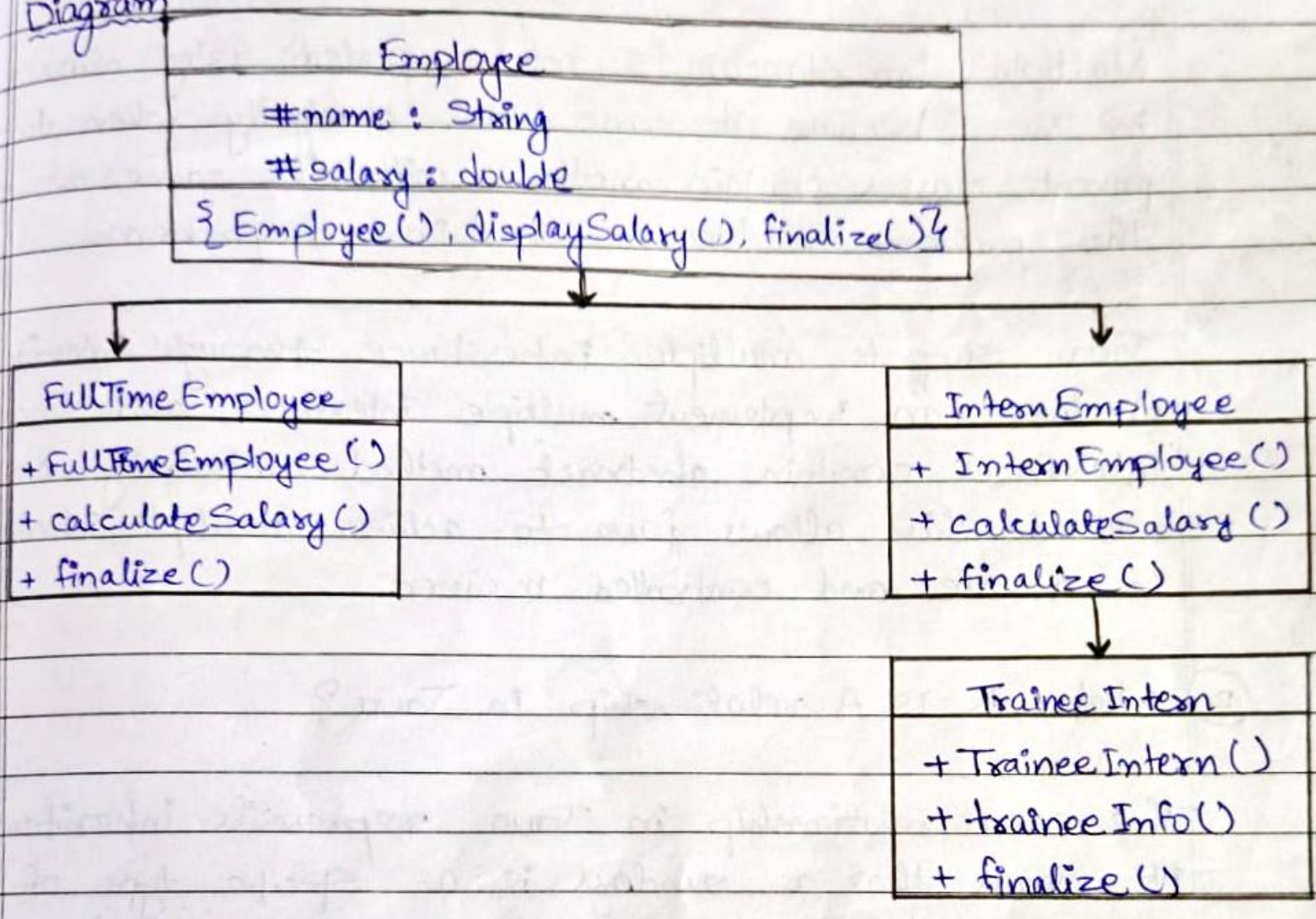
Java supports the following types of inheritance.
- Single Inheritance
- Multilevel Inheritance
- Hierarchial Inheritance
- Multiple Inheritance (through interface only)
- Hybrid Inheritance (through interfaces)

Algorithm

1. Start
2. Create a base class Employee with salary variable and displaySalary() method.
3. Create derived class FullTime Employee extending Employee
4. Create derived class Intern Employee extending Employee.

5. Implement calculate Salary() method in both derived classes.
6. Apply salary hike
   - Full Time Employee → 50% hike.
   - Intern Employee → 25% hike
7. Display salary before and after hike.
8. Stop.

Diagram



Conculsion :

Thus, we have successfully implemented usage of inheritance in Java.

FAQs

(1) Is multiple inheritance supported in Java? How is it achieved?

Multiple inheritance is not supported using classes in Java because it can cause ambiguity when two parent classes contain methods with the same name. The problem is known as Diamond problem.

Java supports multiple inheritance through interfaces. A class can implement multiple interfaces, and since interfaces contain abstract method ambuiguity is avoided. This allows java to achieve multiple inheritance in a safe and controlled manner.

(2) What is IS-A relationship in Java?

The IS-A relationship in Java represents inheritance. It means that a subclass is a specific type of the superclass and can access the properties and methods of the parent class. The relationship is established using the extends keyword.

ex:-
Full time Employee Is-A Employee.
This means a Full Time Employee inherits the features of the employee class.

③ Are constructor and instance initialization block inherited to Subclass?

NO, Constructors and instance initialization blocks are not inherited by Subclasses in Java. Each class has its own Constructors and initialization blocks.
However, a Subclass can invoke the Constructor of its Superclass using the Super ( ) keyword to initialize the inherited data members.